

YERBOGACHEN, Russian Federation

WMO: 248170

Lat: 61.276N

Lon: 108.032E

Elev: 954

StdP: 14.20

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-54.4	-49.7	-57.8	0.2	-54.1	-53.2	0.2	-49.3	14.2	-3.9	12.2	-3.0	0.3	50	0.692

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	84.8	64.7	80.9	62.9	77.1	61.4	66.7	79.3	64.8	77.0	63.1	74.4	5.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	87.7	70.9	60.3	81.0	69.0	58.6	76.0	67.5	31.9	79.8	30.5	77.0	29.1	74.9	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.7	10.1	DB	-57.6	91.0	4.3	2.4	-60.6	92.7	-63.1	94.1	-65.5	95.4	-68.6	97.2	(4)
(5)			WB	-57.5	70.1	4.2	2.3	-60.6	71.7	-63.0	73.0	-65.4	74.3	-68.5	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.6	-20.6	-14.6	5.9	25.9	42.1	59.1	62.7	56.3	42.1	23.6	-5.5	-20.0		
	DBStd	32.78	16.76	15.18	15.45	10.65	7.95	7.54	6.34	6.63	7.88	11.73	18.18	18.59		
	HDD50	11297	2187	1809	1366	723	267	13	1	21	256	819	1666	2169		
	HDD65	15918	2652	2229	1831	1173	710	205	118	279	687	1284	2116	2634		
	CDD50	938	0	0	0	0	22	285	396	215	20	0	0	0		
	CDD65	85	0	0	0	0	0	28	49	8	0	0	0	0		
	CDH74	1513	0	0	0	0	5	572	737	195	4	0	0	0		
	CDH80	418	0	0	0	0	0	170	201	47	0	0	0	0		
(14) Wind	WSAvg	4.0	2.9	3.2	4.1	5.0	5.4	4.4	4.2	3.9	4.2	4.4	3.3	2.7		
(15) Precipitation	PrecAvg	13.3	0.6	0.4	0.5	0.7	1.1	1.8	1.8	2.0	1.4	1.3	1.0	0.7		
	PrecMax	18.7	1.5	1.2	1.3	1.3	2.4	5.0	4.7	5.5	3.0	3.3	2.3	1.7		
	PrecMin	6.8	0.1	0.0	0.1	0.2	0.2	0.1	0.2	0.2	0.2	0.4	0.4	0.2		
	PrecStd	2.6	0.3	0.2	0.3	0.3	0.6	0.9	1.0	1.3	0.7	0.6	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	27.2	45.8	56.4	73.7	89.7	89.9	85.7	72.5	50.6	34.8	26.2		
		MCWB	18.6	25.1	35.5	42.8	54.7	64.7	66.9	66.1	56.1	43.5	32.7	25.1		
	2%	DB	14.1	21.5	38.3	51.5	67.8	85.0	85.1	79.1	65.1	45.4	31.8	21.3		
		MCWB	13.1	19.5	31.1	39.7	50.0	63.7	65.4	62.5	52.4	40.5	30.1	20.5		
	5%	DB	9.4	15.6	33.1	47.3	63.8	80.5	81.7	74.6	60.1	41.1	28.3	14.4		
		MCWB	8.7	14.0	28.2	37.1	48.1	61.4	64.3	60.2	50.7	37.3	26.8	13.6		
	10%	DB	4.9	9.6	29.3	42.8	59.0	76.3	78.0	70.7	55.5	37.4	22.7	8.3		
		MCWB	4.3	8.5	25.2	34.7	45.5	59.8	62.7	58.7	48.5	34.7	21.3	7.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	25.2	36.5	44.0	55.5	67.8	70.1	66.9	58.1	44.4	33.3	25.6		
		MCDB	19.6	26.7	44.6	53.7	69.9	80.7	83.7	81.1	67.3	49.2	34.3	26.4		
	2%	WB	13.1	19.8	31.9	40.4	51.9	65.3	67.8	64.0	54.5	40.9	30.2	20.5		
		MCDB	13.9	21.3	37.5	49.7	63.4	79.6	80.2	75.6	62.3	45.2	31.7	21.5		
	5%	WB	8.7	14.0	28.5	38.0	49.0	63.6	66.0	62.0	51.9	37.9	26.6	13.5		
		MCDB	9.3	15.5	32.9	46.0	61.3	76.8	77.4	71.8	58.8	40.5	28.4	14.2		
	10%	WB	4.2	8.6	25.4	35.4	46.3	61.7	64.2	60.0	49.2	34.9	21.4	7.7		
		MCDB	4.9	9.5	29.3	42.0	57.9	73.4	75.3	68.5	54.8	37.2	22.6	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	24.2	29.0	22.4	19.8	23.4	22.1	21.8	17.8	13.5	17.9	16.0		
		MCDBR	17.3	21.3	25.4	25.6	30.4	32.5	30.5	30.8	27.3	16.9	16.0	15.7		
		MCWBR	16.7	19.9	20.0	16.2	16.0	13.8	13.5	15.4	16.0	12.2	15.3	15.4		
		MCDBR	17.1	21.4	24.6	21.7	26.3	27.3	25.6	25.3	23.7	14.6	16.4	15.7		
	5% WB	MCWBR	16.6	19.9	19.8	14.2	15.2	12.8	12.8	13.9	15.3	11.3	15.7	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.238	0.279	0.306	0.336	0.364	0.394	0.456	0.421	0.332	0.293	0.235	0.205		
	taud		1.945	2.016	2.052	2.103	2.295	2.332	2.133	2.196	2.445	2.287	1.980	1.874		
	Ebn at Noon		141	212	252	269	272	264	241	239	249	213	141	97		
	Edh at Noon		14	25	35	40	36	36	43	37	24	19	13	9		
(44) All-Sky Solar Radiation	RadAvg		128	429	953	1527	1672	1836	1660	1268	772	346	164	66		
	RadStd		7	28	53	95	107	140	119	97	68	40	16	4		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) +4.12	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	+4.12	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page